

1. PLANET DATABASE

AIM

To write a Prolog program to create a Planet Database and retrieve planet information using Prolog queries.

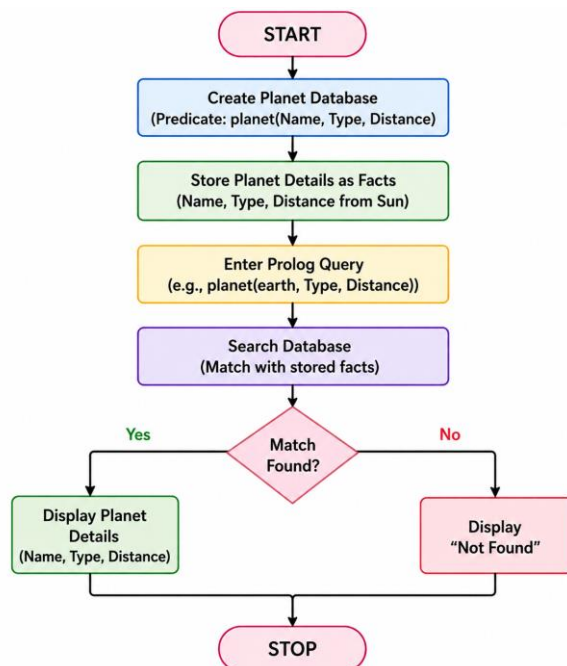
OBJECTIVE

- To create a database of planets using Prolog facts.
- To store the name, type, and distance of planets from the Sun.
- To retrieve planet details using queries.
- To understand database searching in Prolog.

ALGORITHM

1. Start.
2. Define the predicate planet(Name, Type, Distance).
3. Store the planet details as facts.
4. Enter a Prolog query.
5. Search and match the required information in the database.
6. Display the matching planet details.
7. Stop.

FLOWCHART:



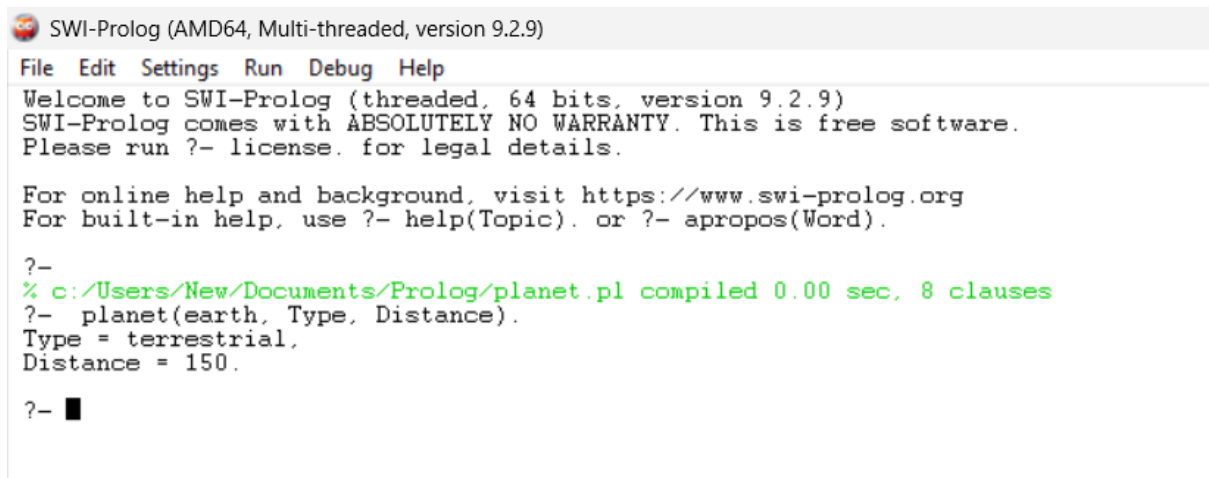
CODE:

```
planet(mercury, 1).  
planet(venus, 2).  
planet(earth, 3).  
planet(mars, 4).  
planet(jupiter, 5).  
planet(saturn, 6).  
planet(uranus, 7).  
planet(neptune, 8).
```

Query

```
?- planet(earth, Position).  
?- planet(Planet, 5).
```

OUTPUT:



```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)  
File Edit Settings Run Debug Help  
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)  
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.  
Please run ?- license. for legal details.  
  
For online help and background, visit https://www.swi-prolog.org  
For built-in help, use ?- help(Topic). or ?- apropos(Word).  
  
?-  
% c:/Users/New/Documents/Prolog/planet.pl compiled 0.00 sec, 8 clauses  
?- planet(earth, Type, Distance).  
Type = terrestrial,  
Distance = 150.  
  
?- ■
```

RESULT:

Thus, the Prolog Planet Database was successfully created, and the required planet information was retrieved using Prolog queries.